Summary of GIT

Q. What is mean by version control System?
Q. What is purpose of version control System?
Q. What are different types of Version Control Systems?
Q. What are the key benifits of Version Control System?
Q. What is difference between Centralized Version Control System and Distributed Version Control System?
Q. Which version control system google uses.?
Q. Which control system AWS uses?

Q. What is GIT?
-> GIT is a free and open source distributed version control tool.

Q. What is GITBASH?
-> GITBASH is a command line interface for windows users to interact with git.

Q. What is GITHUB?
-> GITHUB is web-based git repository.

Q. How to install git on windows operating System?
-> Download gitbash and install gitbash.
gitbash internally installs git.

Q. How to install git on Linux operating System?
-> On Linux operating system GIT is installed through the command.
Steps: open terminal and execute below command

```
sudo apt install git
```

here sudo stands for Super User Do

Q. Why gitbash is required for windows operating system but not for Linux operating System?

-> Git is developed in C language and runs on Linux operating system.
Linux has inbuilt linux environment but windows don't have.
So to get Linux environment gitbash is required on windows.

Q. git is which type of version control system, centralized or distributed?

Q. In which language git is developed?

Q. Who is maintaing git software versions?

Q. What is difference between git and Gitbash?

-> GIT is version control tool where as GITBASH is command line interface for windows users to work on GIT.

Q. what is difference between git and GitHub?

-> GIT is version control tool where GitHub is web-based git repository.

Q. What is difference between GitHub and GitLab?

-> Both are same but GitLab offers more features.

Q. How you cross check git is installed or not your system?

Q. How you cross check installed git version?

Q. Which command is used to check the status of local repository?

Q. What is mean by Untracked files?

-> Newly added files are considered as Untracked Files

-> On gitbash,Untracked file names are shown in read color

Q. What is mean by staged or tracked files?
-> The files present in staging area are considered as staged file.
-> On gitbash,tracked file names are shown in green color

Q. What is mean by modified files?
-> If we change the content of tracked file(staged or committed) then it is considered as modified file.
-> On gitbash,modified file names are shown in read color

Q. What is mean by deleted files?
-> If we delete tracked file(staged or committed) then it it is considered as deleted file.
-> On gitbash,deleted file names are shown in read color

Q. What is mean by branch in git?
Q. Which branch is considered as default branch or initial branch in git?
-> master or main is considered as default branch or initial branch in git

Q. What are different levels(scope) of Configuration?
Q. What is mean by System Level Configuration?
Q. What is mean by Local Level Configuration?
Q. What is mean by Global Level Configuration?
Q. System Level Configuration file is present in which location?
Q. Local Level Configuration file is present in which location?
Q. Global Level Configuration file is present in which location?

Q. How to add remote origin in local repository?
-> To add remote origin use below syntax:
Syntax: git remote add origin repository_url

Q. When we need to add remote origin in local repository?
-> You have created repository on local system and you want to push local repository on github,that time we need to add remote origin in local repository.

Creating Local repository and push on github Steps
-> 1. Create an empty directory on local system
-> 2. Initialize this empty directory as git repository by using "git init" command.
-> 3. Add some dummy files into repository
-> 4. add this file into staging area by using "git add" command
-> 5. add this file into staging area by using "git commit" command
-> Create same repository on github

-> 7. Create link between local repository and remote repository. i.e add remote origin
Syntax: git remote add origin repository_url
->8. Set up stream branch to push your changes
Syntax: git push -u "remote_name" "remote_branch_name"
Example: git push -u origin main
ones this command is executed, local repository get push on github
Note
Local repository initial branch name and remote repository initial branch name must be same. If name is not same then rename local repository branch name by using "git branch -m remote_branch_name"

Q.How to cross check parent Branch Name?
-> use "git reflog" command
-> Syntax: git reflog show branch-name

Q. When we execute "git log" command then which information get display on the screen?
-> When we execute the "git log" command then below information get display on the screen

1) commit hash(commit id, it is 40 character long)
2) Author (Who has committed the changes)
3) Date and Time(When committed)
4) Commit message

Q. How you exit from "git log" command?
-> To exit from "git log" command press "q"(quite)

Q. What is use of git stash command?
-> git stash command is used to save the changes temporarily.

Q. What is use of .gitignore file?
-> To specify which files and directory should not be tracked by GIT.
i.e To exclude the files and directory from tracking.

-> IMP Note: Files specified in the ".gitignore" should not be tracked but GIT keep track of ".gitignore" file.
i.e: GIT maintain versions of ".gitignore" file.

Q. How many ".gitignore" files can have for a repository?
-> multiple

Q. Which symbol is used to comment the code in GIT?
-> Hash symbol is used to comment the code.

Some examles for ignoring the File
-> # 1) Ignore F1.txt
F1.txt
2) Ignore all text files
*.txt
3) Ignore all log files
*.log
4) Ignore Abc named directory
Abc
5) Ignore all text files except F10.txt
*.txt
!F10.txt
6) Ignore specific files from directory
Abc/F5.txt

git internals:about .git folder

Q. When .git folder get created?
-> When we clone a repository by using "git clone" command or when we initialize git repository by using "git init" command then internally .git folder get created.

Q. Why .git folder is hidden from user?

Q. HEAD reference is stored in which file?
-> HEAD reference is stored in "HEAD" file.

Q. HEAD always point to which branch?
-> HEAD always point to current branch

Q. Repository specific configuration settings available in which file?

-> Repository specific configuration settings available in /.git/config file

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Q. index file holds which information?

-> index file holds file path,file permission and object IDS of staged files.

Q. FETCH_HEAD file holds which information?

-> FETCH_HEAD file holds latest fetched changes from remote.

-> FETCH_HEAD pointer point to that changes.

-> If we want merge latest changes in local branches then use "git merge FETCH_HEAD" command.

-> If we want to checkout fetched changes then use "git checkout FETCH_HEAD"

Q. ORIG_HEAD file holds which information?

-> ORIG_HEAD holds previous HEAD reference(address)

#. In GIT we have three pointer HEAD,FETCH_HEAD and ORIG_HEAD

1) HEAD: Holds current branch reference
2) FETCH_HEAD: Holds latest commits fetched from remote
3) ORIG_HEAD: Holds previous HEAD reference

Q. logs directory holds which information?

-> logs directory holds commit history and branch history

Q. blob object holds which information?
Q. tree object holds which information?
Q. commit object holds which information?
Q. exclude file present in which directory?
Q. What is use of exclude file in GIT?
Q. What is purpose of git configuration file?
Q. In git configuration file user name and email present under which section?
Q. What is difference between git pull and git fetch?
Q. What is mean by conflict?
Q. Why conflict came while merging changes of one branch into another branch ?